

Identification

- Identified rooftop units and fan room systems
- Created a unique ID system
- Units are based on unit location instead of quadrant system



Image Capture

- Took pictures of:
- Label
 - Switch
 - Overall unit/compressor/fan
- Added them to an organized PowerPoint for referencing



Data Recording

- From the images, I recorded:
- Manufacturer
 - Refrigerant
 - Model Number
 - Charge
 - Serial Number
 - MFR Date

Allowed for research and location of all other required data

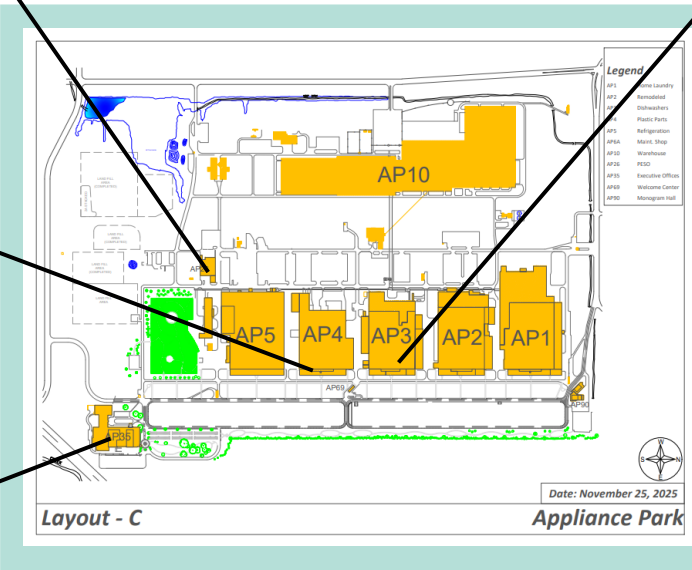


Operational Efficiency

Optimizing HVAC Inventory for the Front Office Maintenance Team

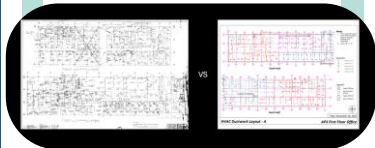
Objective

To streamline HVAC Inventory workflows, enabling the maintenance team to deliver faster, more reliable service while minimizing downtime and resource waste.



CAD Drawings

- Reviewed drawings to locate AHUs
- Some were last updated 50-70 years prior
- Created new AutoCAD files to calculate total charge



Labeling

Created labels with all key information for each unit

- White = under 15 lbs of refrigerant
- Orange = 15-49 lbs of refrigerant
- Purple = 50+ lbs of refrigerant



Importance

After time study:

- 20 HVAC complaints/wk
- Cuts resolution time by **74%**
 - From 2.5-4.5 hours to 45 minutes
- Cuts resolution cost by **84%**
 - From **\$191.29** per complaint to **\$30.34** per complaint

	Old Method	New Method	% Reduction
Time Spent per Complaint (hours)	3.17	0.82	
Combined Time Spent per Week (hours)	62.4	16.4	74%
Combined Time Spent per Month (hours)	251.6	65.6	
Combined Time Spent per Year (hours)	3019.2	787.2	
Cost per Complaint	\$191.29	\$30.34	
Combined Cost per Week	\$525.80	\$50.60	84%
Combined Cost per Month	\$1,912.90	\$307.20	
Combined Cost per Year	\$22,954.80	\$3,686.40	
TOTAL TIME SAVED		2250 hours per year	
TOTAL COST SAVED		\$154512 per year	